

Sericulture

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Sericulture or silk production is the breeding and management of silkworms for the commercial production of silk.

It includes:

- Cultivation of food plants like mulberry to feed silkworms
- Rearing of silkworm larvae
- Extraction and processing of silk fibers from cocoons

Sericulture is a **farm-based, labour-intensive, income-generating agro-industry** that links both agriculture and industry.

Silk

Silk, known as “*The Queen of Textiles*”, represents luxury, elegance, and comfort.

- Chemically, silk is a protein fiber secreted in liquid form by a silkworm caterpillar.
- The silkworm feeds on selected food plants and spins a cocoon as a protective shell.
- The life cycle of silkworm: **egg** → **caterpillar (larva)** → **pupa** → **moth**.
- Man interrupts this cycle at the cocoon stage to obtain silk filament used for weaving fabric.

Global Importance

- Asia produces over **95%** of the world’s silk.
- **China** – leading producer and exporter.
- **India** – second-largest producer and the largest consumer of silk.
- **Major Indian states for mulberry silk:** Karnataka, Andhra Pradesh, Tamil Nadu, Jammu & Kashmir, West Bengal.

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- **Non-mulberry silk states:** Jharkhand, Chhattisgarh, Odisha, and North-Eastern states.

+ Types of Silk

There are **five major commercial types of silk** obtained from different species of silkworms feeding on various host plants:

- **Mulberry**
- **Tropical Tasar**
- **Oak Tasar**
- **Muga**
- **Eri**

India produces **all four natural varieties** – Mulberry, Tasar, Muga, and Eri.

The non-mulberry silks (Tasar, Muga, Eri) are called **Vanya silks** or **wild silks**.

Colour of Different Silks

Type	Colour
Mulberry	White / Cream / Yellow / Green
Tropical Tasar	Copperish
Eri	Creamy white
Muga	Golden yellow

India is the **only country in the world** that produces all **four major natural varieties** of silk — Mulberry, Tasar, Muga, and Eri.

The non-mulberry silks (Tasar, Muga, and Eri) are also called **Vanya silks** or **wild silks** because they are obtained from silkworms that feed on forest trees and are mostly reared outdoors.

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1. Mulberry Silk

- Produced by the **mulberry silkworm (*Bombyx mori*)**, which is completely domesticated.
- It feeds only on the leaves of the **mulberry plant (*Morus alba*)**.
- The silk is **fine, smooth, lustrous**, and forms the bulk of India's total silk production.
- The cocoons are **white, cream, or yellowish-green** in color.
- The life cycle is completed under controlled indoor conditions.
- Major producing states: **Karnataka, Andhra Pradesh, Tamil Nadu, West Bengal, Jammu & Kashmir.**

Uses:

Mulberry silk is used for high-quality textiles such as sarees, garments, upholstery, and export-quality fabrics.

2. Tasar Silk (Tropical Tasar)

- Also called **Tussah silk**.
- Produced by the **wild silkworm *Antheraea mylitta***.
- These silkworms are **semi-domesticated** and reared in open forests.
- They mainly feed on the leaves of **Asan (*Terminalia tomentosa*)** and **Arjun (*Terminalia arjuna*)** trees.
- Tasar silk is **coarse, strong, and has a natural copperish brown color**.
- Mainly produced in **Jharkhand, Chhattisgarh, Odisha, Maharashtra, Madhya Pradesh, and Andhra Pradesh.**

Uses:

Tasar silk is used for **furnishings, interiors, and decorative fabrics** due to its coarse texture and natural sheen.

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3. Oak Tasar Silk

- Produced by the **silkworm *Antheraea proylei***.
- It is a **finer variety of Tasar silk**, compared to tropical Tasar.
- The worms feed on the **oak tree (*Quercus species*)**, found abundantly in **Himalayan and sub-Himalayan regions**.
- The silk is **soft, lighter in color**, and has better texture and feel than tropical Tasar.
- It is mainly produced in **Himachal Pradesh, Manipur, Nagaland, and Arunachal Pradesh**.

Uses:

Oak Tasar silk is used for **premium fabrics, scarves, and lightweight clothing**.

4. Muga Silk

- Produced by the **silkworm *Antheraea assamensis***, which is endemic to **Assam**.
- The silkworms feed on the leaves of **Som (*Machilus bombycina*)** and **Soalu (*Litsaea polyantha*)** plants.
- Muga silk is known for its **natural golden-yellow luster** and **exceptional durability**.
- The silk becomes **more glossy with each wash** and is considered a **heritage product** of Assam.
- Muga is a **wild silk**, reared outdoors in natural conditions.

Uses:

Muga silk is used for **traditional Assamese attire (like Mekhela Chador)** and **royal garments**.

5. Eri Silk

- Also known as **Endi silk** or **Ahimsa silk** because the pupae are not killed during processing.

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- Produced by the **silkworm** *Philosamia ricini*.
- The worms feed mainly on **castor leaves** (*Ricinus communis*).
- The cocoons are **open-ended**, allowing the moth to emerge naturally.
- The silk is **woolly, soft, and warm**, with a **creamy white color**.
- Eri silkworms are **completely domesticated** and reared indoors.
- Mainly found in **Assam, Meghalaya, Nagaland, and other North-Eastern states**.

Uses:

Eri silk is used for **shawls, quilts, and winter garments** due to its warmth and texture.

Silk Type	Silkworm (Scientific Name)	Colour	Host Plant	Region
Mulberry	<i>Bombyx mori</i>	White/Cream/Yellow/Green	Mulberry	All India
Tropical Tasar	<i>Antheraea mylitta</i>	Copperish	Asan, Arjun	Central & South India
Oak Tasar	<i>Antheraea proylei</i>	Light brown	Oak	Himalayan, NE India
Eri	<i>Philosamia ricini</i>	Creamy white	Castor	NE India
Muga	<i>Antheraea assamensis</i>	Golden yellow	Som, Soalu	Assam, NE India

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Classification

Based on Occurrence:

- **Domesticated:** Mulberry, Reared Tasar
- **Wild (Vanya):** Tasar, Muga, Eri

Based on Life Cycles per Year:

- Univoltine
- Bivoltine
- Multivoltine

Voltinism

Voltinism is the ability of silkworms to produce one or more generations in a year.

Type	Features	Example
Univoltine	One generation per year; long larval period; high-quality silk; adapted to cooler regions	European races, Oak Tasar
Bivoltine	Two generations per year; larvae robust; good cocoon quality	Japanese and Chinese races
Multivoltine	More than three generations per year; short life cycle; small cocoons; poor silk quality	Indian races

Comparative points:

- Bivoltine eggs can be stored 6–10 months; multivoltine only ~20 days.
- Bivoltine filament: 1000–2600 m; multivoltine: 400–500 m.
- Bivoltine less heat-tolerant (<28°C); multivoltine can tolerate higher temperatures.

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Moulting and Moulting

Moulting:

Silkworms stop feeding and shed their old skin to grow larger. Occurs 4–5 times in a 30–35-day life span.

Moulting:

Classification based on number of moults – tri, tetra, penta, or hexa-moults.

Life Cycle of Mulberry Silkworm (*Bombyx mori*)

Duration: **45–55 days**

Stages: **Egg → Larva → Pupa → Moth**

Stage	Duration	Description
Egg	9–10 days	Laid by female moth; hatches in 7–10 days under 23–28°C, 85–90% humidity
Larva	24–28 days	Feeds on mulberry leaves; five instars, four moults
Pupa (Cocoon)	8–10 days	Larva spins continuous silk thread forming cocoon
Moth	3–4 days	Emerges, mates, lays eggs; life cycle repeats

Processing of Silk

1. **Stifling:** Killing pupae by heat or steam.
2. **Reeling:** Unwinding silk filament from softened cocoons.
3. **Twisting:** Combining strands into yarn.
4. **Degumming:** Removing sericin by boiling cocoons.
5. **Weaving:** Converting silk yarn into fabric.
6. **Deflossing:** Removing rough outer silk to improve quality.

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Diseases of Silkworm

Disease	Causative Agent	Type	Key Symptoms
Pebrine	<i>Nosema bombycis</i>	Protozoan/Fungal	Clumped eggs, curled wings, irregular larvae
Flacherie	<i>Streptococcus</i> , <i>Bacillus</i> , or virus	Bacterial/Viral	Larvae soft, vomit gut juice, foul smell
Grasserie	<i>Bombyx mori</i> NPV	Viral	Shiny skin, swelling, rupture, white haemolymph
Muscardine	<i>Beauveria bassiana</i> (white), <i>Metarrhizium anisopliae</i> (green)	Fungal	Inactive larvae, white fungal coating after death